GIACOMO MARCIANI

SENIOR SOFTWARE ENGINEER @ Amazon Web Services

Building High Performance Computing in the Cloud

LOCATION

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CONTACTS

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INTRODUCTION

I am a Senior Software Engineer with 8+ years at AWS, specializing in High Performance Computing and GPU acceleration infrastructure in the cloud. I own the full lifecycle end to end, from strategy to design, implementation, DevOps, and on-call operations, with experience as a technical leader and supporting product development decisions.

I have contributed to AWS ParallelCluster, NICE EnginFrame, Elastic Graphics, and Elastic Inference. My background spans software engineering, distributed systems and big data. Before AWS, I founded the first ACM Student Chapter in Rome, co-founded a startup, and taught computer science at university. I also led research projects that resulted in publications in scientific journals and presentations at international conferences.

What makes me a good engineer and a good leader is my curiosity and the constant desire to keep getting better in how I build, how I lead, and how I show up for my team.

WORK EXPERIENCE

Sr Software Engineer
Amazon Web Services, 2021-Present

Technical leader of AWS ParallelCluster. My responsibilities span the software lifecycle end to end: strategy, design, implementation, ops, oncall activities, mentoring and hiring.

Software Engineer II
Amazon Web Services, 2019-2021

Starting in 2021, I took on AWS ParallelCluster, where I became the de facto technical leader of the team. My key contributions include Multi-User environment, Dynamic File System Mounting, Support for GPU intensive instance types, enabling ParallelCluster in US isolated regions, Terraform integration, ParallelCluster UI, serving as Security Guardian, driving roadmap and releases and resolving high-severity escalations for key customers. I also led the complex shutdown of Elastic Graphics and Elastic Inference, navigating deep integrations with EC2/ECS.

Software Engineer I
Amazon Web Services, 2018-2019

Worked on Elastic Graphics and Elastic Inference, whose goal was to attach cost-effective GPU acceleration to EC2 instances and other AWS services. I worked on the launch of Elastic Inference at Re:Invent 2018 and owned the component to measure the consumption and billing of computational resources.

R&D Intern
Accenture, 2017-2018

Designed solution for the elastic auto-scaling of containerized applications using Reinforcement Learning in Kubernetes.

Chairman, Founder
ACM Rome Tor Vergata, 2016-2018

Founded the ACM Student Chapter of University of Rome Tor Vergata, the first one in the city. Managed the team and promoted interaction between academia and industry.

Teaching Assistant
University of Rome Tor Vergata, 2016-2018

Tutored BSc students and evaluated their theoretical and practical exams for courses in Foundations of Computer Science and Algorithms Engineering at the BSc in Computer Science.

Management Engineer, Co-Founder
Sbaam.com, e-Commerce, 2011-2012

Co-founded the startup, bringing it as finalist during the InnovActionLab startups contest. Responsible for the business plan development and pitches in front of hundreds investors.

Mathematics Tutor
Freelance, 2011-2017

Gave private lessons of math to students from secondary and high schools. My greatest achievement was that many of them became passionate about math and enrolled in engineering.

MAIN PROJECTS

AWS ParallelCluster and ParallelCluster UI
Amazon Web Services, 2021-Present | HPC

Open source tool to deploy and manage HPC clusters in the cloud, with millions dollars in revenues and very big customers.

NICE EnginFrame
Amazon Web Services, 2021 | HPC

Web application to submit HPC workloads to both on-premises and AWS clusters.

AWS Elastic Graphics and Elastic Inference
Amazon Web Services, 2018-2023 | Web Service

Services to attach GPU-powered acceleration for OpenGL and ML workloads to EC2, ECS and Sagemaker to reduce the cost of running graphics and machine learning applications.

Crime Graph
Big Data, IEEE FiCloud, 2017 | Java, Kafka, Flink, Neo4j

Data stream processing application for the real-time analysis of evolving criminal networks, uncovering potential and hidden relationships.

NNML-HPC
High Performance Computing, 2017 | C, CUDA

Deep neural network for hand-written digits recognition, implemented with CUDA/C and OpenMP.

OntoQA
Artificial Intelligence, 2017 | NLP, Java, Protege

Question&Answer web application leveraging ontology-driven natural language processing.

Godot
App for Transports, 2013 | Java, Android

Mobile app to solve discomforts caused by illegal parking, leveraging QRCodes and NFC.

See more on my GitHub

PUBLICATIONS

A data streaming approach to link mining in criminal networks
Proceedings of the 5th IEEE International Conference on Future IoT (FiCloud 2017)

Real-time analysis of social networks leveraging the Flink framework
Proceedings of the 10th ACM International Conference on Distributed and Event-Based Systems (DEBS 2016)

Marciani Normal Form of context-free grammars
Arxiv (2016)

EDUCATION

MSc Software Engineering and Data Science
University of Rome Tor Vergata, Rome, IT
2015-2018 Grad Avg 29/30 - GPA 4/4 *

BSc Software Engineering
University of Rome Tor Vergata, Rome, IT
2011-2015 Grad 110/110 cum laude

BSc Management Engineering
University of Rome Tor Vergata, Rome, IT
2009-2011 Interrupted

Cambridge Summer School
Girton College, Cambridge, UK
2005, 2006 Grad English Proficient

Classical License
Liceo T. Tasso, Rome, IT
2004-2009 Grad 77/100

MAIN COURSES

Foundation of Computer Science	30L
Software Engineering	30L
Internet Engineering	30L
Web Programming	30L
Web Algorithms	30L
Big Data Architectures	30L
Artificial Intelligence	30L
Distributed Systems	30L
Computer Security	30L
Computer Architectures	30
Optimization Methods	29
Mobile Programming	29
Algorithms Engineering	29
Multi Core Programming	29
Performance Modeling	29
Operations Research	27
Automata and Languages	26
Economics	30
ICT Project Analysis	30

HONOURS

Lecturer at Master CESMA
University of Rome Tor Vergata, 2016

Finalist at ACM DEBS Grand Challenge
10th ACM International Conference on Distributed Event Based Systems, 2016

Speaker at InnovAction Lab Grand Challenge
LVenture Groups, 2013

Best Student in Computer Architectures
University of Rome Tor Vergata, 2012

Alumno of InnovAction Lab
LVenture Groups, 2011

Counselor of Engineering
University of Rome Tor Vergata, 2010

MAIN SKILLS

Cloud Amazon Web Services, Digital Ocean, Railway

HPC ParallelCluster, EnginFrame, Slurm, Nvidia GPUs, Cuda, MPI, Lustre, EFA

Big Data Flink, Hadoop Distributed File System, Hadoop MapReduce, Pig, Hive, Kafka, Flume, Elasticsearch, Kibana, Grafana, Kubernetes

Web Spring, Nginx, Tomcat, React, NodeJS, HTML, CSS, Sass, Postman, Hugo, Wordpress

Automation CloudFormation, Terraform, Chef

Data SQL, NoSQL, Graph, Caching

Languages Python, Java, Kotlin, Ruby, Javascript, C, Cuda, Bash, HTML, OWL, SPARQL, AMPL, MIPS, Latex

Platforms Linux, OSX, Docker

Applied Math Matlab, AMPL, IBM Cplex

Process Agile, SCRUM

LANGUAGES

Italian
Native

English
Proficient

Maximum is 30/30. L stands for cum laude. Grade 27 is equivalent to A. * Thesis pending. I hereby authorize to use and process my personal details contained in this document (Italian Decree 196 30/06/2003).